MIL-T-37638A
24 February 1978
SUPERSEDING
MIL-T-37638(DSA-DM)
28 September 1976

MILITARY SPECIFICATION

TEST TUBE, 12 by 75 mm, POLYSTYRENE, DISPOSABLE

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers clear 12 x 75 millimeter (mm.) lipless polystyrene plastic test tubes suitable for laboratory purposes.

2. APPLICABLE DOCUMENTS

* 2.1 Issues of documents. The following documents, of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

PPP-B-566	- Box, Folding, Paperboard.
PPP-B-585	- Boxes, Wood, Wirebound.
PPP-B-601	- Boxes, Wood, Cleated-Plywood.
PPP-B-621	- Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636	- Boxes, Shipping, Fiberboard.
PPP-B-676	- Box, Setup.

MILITARY

MIL-L-10547	- Liners, Case and Sheet, Overwrap; Water-Vaporproof, or Waterproof, Flexible.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Defense Personnel Support Center, Directorate of Medical Materiel, DPSC-ATT, 2800 South 20th Street, Philadelphia, PA 19101, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 6640

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STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

(Copies of specifications, standards, drawings and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Material. The test tubes shall be fabricated of clear polystyrene plastic.

* 3.2 Design and construction. The test tubes shall be cylindrical with closed rounded bottoms. Rounded bottoms with a recess on exterior surface and a corresponding nipple on inside surface, to eliminate gate stub protrusion beyond end at test tube, shall be acceptable. The open ends shall be without lips. The dimensions of the test tubes shall conform to table I.

TABLE I. Dimensions of test tubes.

Characteristic	Dimension (mm.)
Overall length	75.0 + 1.5/-1.0
Outside diameter at center of tube	12.5 max.
Inside diameter at open end	9.5 min.

The overflow capacity of the test tubes shall not be less than 4.0 milliliters (ml.).

* 3.3 Workmanship. Workmanship shall be first class throughout. The test tubes shall be free of defects which detract from their appearance or impair their serviceability. The test tubes shall be symmetrical in shape, and free of sharpness, distortion, chips, and cracks. The test tubes shall be clear, dry, ready for use, and free from grease, dirt, stains, and other foreign matter. The test tubes shall be sufficiently free of clusters of scratches and abrasions so as not to interfere with the transparancy of the plastic.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification to assure supplies and services conform to prescribed requirements.

4.1.1 Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of three years after delivery of the supplies to which such records relate.

4.1.2 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual and auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

4.2 Sampling.

4.2.1 For examination. Sampling for examination shall be conducted in accordance with table II and MIL-STD-105. The unit of product for examination shall be an individual test tube. No more than 10 test tubes shall be selected from a package of 250 tubes (see 5.1.1.1.1).

TABLE II. Sampling for examination.

	Inspection level	AQL (percent defective)
For visual examination		
Major defects	II	1.0
Minor defects	II	2.5
For dimensional examination	S-3	2.5

4.3 Examination. The test tubes shall be examined to determine compliance with the requirements contained in this specification.

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* 4.3.1 Classification of defects. Examination shall be conducted in accordance with table III, but shall not be restricted to the classified possible defects listed in the table.

TABLE III. Classification of defects.

Categories	Defects
Major	
101	Test tube not free of chips, cracks, or tears.
102	Test tube not free of sharpness that tends to cut the hand.
103	Test tube not free of physical distortion.
104	Test tube not free of hazy appearance that obstructs transparancy of plastic material.
105	Test tube not free of clusters of scratches or clusters of abrasions that obstruct transparancy of plastic material.
106	Test tube not dry.
107	Test tube surfaces not free of grease, dirt, stains, or other foreign matter.
Minor	
201	Plastic material of test tube not free of imbedded foreign material.

4.3.2 Dimensional examination. The test tubes shall be examined for defects in dimensions. Any dimension not within specified limits shall be considered as a defect. The capacity of the test tube may be determined volumetrically or by calculation based on interior measurements.

4.4 Packaging inspection. The inspection of the unit package, packing and marking for shipment and storage shall be in accordance with quality assurance provisions of the applicable container specification and the marking requirements of MIL-STD-129.

5. PACKAGING

5.1 Packaging. Packaging shall be level A or C, as specified (see 6.1).

5.1.1 Unit of issue. One package containing 1000 tubes, as specified, constitutes one unit of issue.

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5.1.1.1.1 Unit package. Two hundred and fifty tubes shall be clean, dry and neatly packaged in a snug fitting box conforming to PPP-B-566 or PPP-B-676. Closure of boxes shall be secured by means of tape as specified in the appendix of PPP-B-676. As an alternate, two hundred and fifty tubes shall be packaged in a paperboard tray and shrink-wrapped in plastic film. Four trays or four boxes, each containing 250 tubes, shall be further packaged in a box conforming to PPP-B-636, class domestic. Closure shall be as specified in the appendix of the box specification. Prior to packaging the four shrink-wrapped trays, a fiberboard pad shall be placed on the inside bottom of the box when metal fasteners are used in assembly of the box.

5.1.2 Level C. Units shall be packaged in standard commercial containers of the size and kind commonly used, which will afford the degree of protection required for shipment and use of the product for its intended purpose.

5.2 Packing. Packing shall be level A, B or C, as specified (see 6.1).

5.2.1 Level A. Four unit packages shall be packed in an exterior container designed for a type 2 load and constructed in accordance with PPP-B-585, class 1, style 3; PPP-B-601, overseas type; PPP-B-621, class 2; or PPP-B-636, class weather-resistant. Closure and strapping shall be as specified in the appendix of the applicable box specification.

5.2.1.1 Case liner. Each level A wood box shall be lined with a waterproof case liner conforming to MIL-L-10547. Closure and sealing shall conform to applicable paragraph of appendix thereto. Case liner shall not be required for fiberboard boxes. Each fiberboard box shall be waterproofed in accordance with appendix of PPP-B-636.

5.2.2 Level B. Four unit packages shall be packed in an exterior container designed for a type 2 load and constructed in accordance with PPP-B-585, class 1, style 3; PPP-B-601, domestic type; PPP-B-621, class 1; or PPP-B-636, class domestic. Closure shall be as specified in the appendix of the applicable box specification.

5.2.3 Level C. The subject commodity shall be packed in substantial commercial containers of the type, size and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to point of delivery called for in the contract or purchase order.

5.3 Marking.

5.3.1 Interior packages. Each unit package shall be marked in accordance with MIL-STD-129.

5.3.2 Exterior container. Exterior container shall be marked in accordance with MIL-STD-129.

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6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

a. Title, number and date of this specification.
b. National Stock Number (NSN).
c. Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

6.2 This specification covers the following item as listed in the Federal Supply Catalog:

National Stock Number	Item Identification
6640-00-247-6372	TEST TUBE, 12 by 75 mm. Polystyrene Clear, Disposable, 1000s

6.3 This specification does not cover all types, classes, grades or sizes of the commodity indicated by the title of this specification, or those which are commercially available, but is intended cover the types normally procured to meet Military requirements.

6.4 The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire contract irrespective of the marginal notations and relationship to the last previous issue.

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Project No. 6640-1017

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DOCUMENT IDENTIFIER *(Number)* AND TITLE
MIL-T-37638A TEST TUBE, 12 by 75 mm, POLYSTYRENE, DISPOSABLE

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

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